

# Particulate Matter Monitoring & Health Effects

Issue 2026-07-29 | PubMed entry window 29 July 2026 | 25 records in scope, 3 excluded

**25**RECORDS IN SCOPE  
(3 excluded)**9**SUBTOPIC CLUSTERS  
REPRESENTED**10**INSTRUMENTATION /  
MODELLING RECORDS**8**EFFECT ESTIMATES  
EXTRACTED**4**REVIEWS /  
EVIDENCE SYNTHESSES

## Signal of the day

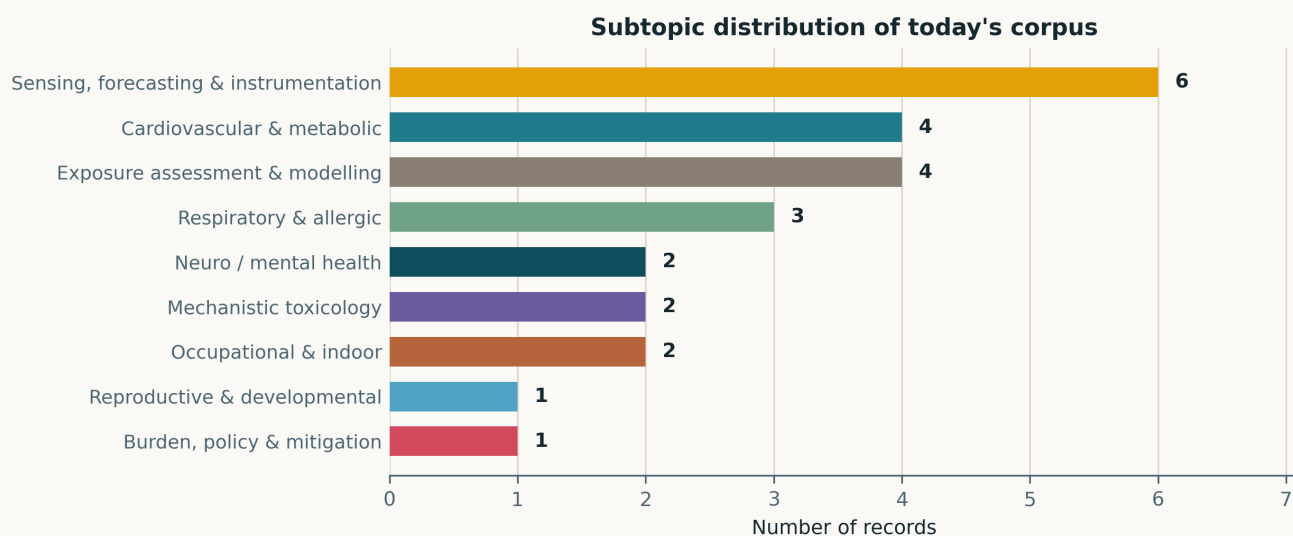
### A US-derived PurpleAir correction does not travel. Region-specific calibration is not a refinement — it is a precondition.

Ginsburg et al. (*Aerosol Sci Technol*) evaluated three widely used US correction equations against **1,013 PurpleAir units co-located within 500 m of reference-grade monitors on six continents** (Jan 2023–Mar 2025). Raw readings overestimate  $PM_{2.5}$  in most regions and fail EPA sensor performance targets outright; the US-derived corrections improve accuracy but **do not transfer outside North America**. A global quadratic model helps in Australia/New Zealand and degrades elsewhere. Only *regional* calibrations consistently met EPA 24-h thresholds, raising 24-h AQI category correctness to  $\geq 77\%$ . Read alongside Feng et al. (*Meas Sci Technol*), whose review finds no universally optimal strategy — ML wins on predictive error ( $< 10 \mu g m^{-3}$ ), physics-based models on interpretability, hybrids on transferability — and the operational conclusion for distributed networks is uncomfortable: **a network without local co-location has an unquantified bias, and no published equation removes it**. The mechanism is not mysterious. Both papers finger RH sensitivity and aerosol-composition variability, which is exactly what a nephelometric scattering-to-mass inversion cannot see.

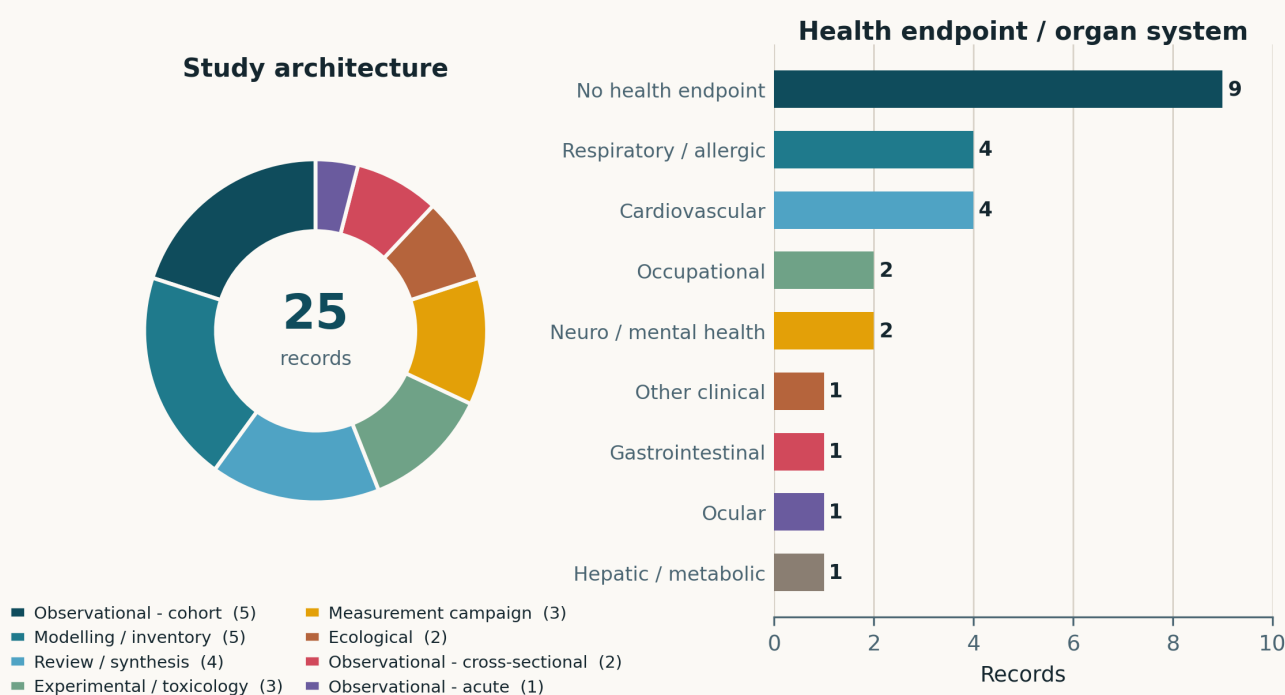
## What changed today

- **Compound-hazard epidemiology arrives with adequate power.** Zhao et al. (*JACC*) analyse **5.9 million CVD deaths across six countries** and find co-occurring heat waves and substantial fire-sourced air pollution carry RR 1.115 (1.100–1.130), above heat-only (1.083) and smoke-only (1.014), with a small but positive additive interaction (RERI 0.018, 95% CI 0.002–0.035). The interaction term is real but modest — most of the compound risk is heat. Resist the synergy headline.
- **Two independent imaging cohorts converge on subclinical coronary structure.** Belay et al. tie cumulative *bushfire-related*  $PM_{2.5}$  to total plaque volume ( $+10.14 mm^3$  per  $1 \mu g m^{-3}$ , 95% CI 1.45–18.82) in 365 serial CCTA participants at a 10-year mean of only  $\sim 5.9 \mu g m^{-3}$ ; Chen et al. tie 1-year  $PM_{2.5}/PM_{10}/BC$  to pericoronary fat attenuation index in 5,528 patients ( $+7.57/8.72/2.35 HU$  per IQR in the RCA). Structural and inflammatory endpoints, same direction, different populations.
- **A well-designed  $PM_{2.5}$  null.** In 1,587 ECHO-PATHWAYS children, infancy  $PM_{2.5}$  and  $NO_2$  showed *no* association with atopic dermatitis at 4–6 years; only  $O_3$  did (aRR 1.07 per 1 ppb, 1.01–1.13), and only in children of lower-educated mothers. Well-powered nulls from validated spatiotemporal models deserve the same shelf space as positives.
- **The satellite-fusion literature has started auditing its own bias.** Ding et al. argue that ground-station clustering in cities systematically inflates retrieved rural  $PM_{2.5}$ ; their urban–rural balanced framework cuts clean-site RMSE from 18.18 to  $8.31 \mu g m^{-3}$  and lowers the national mean by  $\sim 55\%$ , which would substantially revise PM-attributable mortality in low-density areas. If that holds, a large body of exposure–response work inherits an upward exposure bias.
- **Sensor siting became a tractable optimisation.** Patel et al. (AAAI) maximise mutual information over continuous sensor coordinates by gradient descent, with runtime independent of both placement count and candidate-grid density — it approaches greedy MI quality orders of magnitude faster. Directly usable for network expansion planning.
- **Caveat on a headline calibration number.** Msigwa et al. report RMSE  $0.325 \mu g m^{-3}$  and  $R^2$  0.999 for a transformer/TCN hybrid against a reference instrument. That is below the reference's own uncertainty and is not credible as a generalisation estimate; it points to temporally adjacent train/test splits rather than held-out deployment. Treat as an architecture description, not a performance claim.

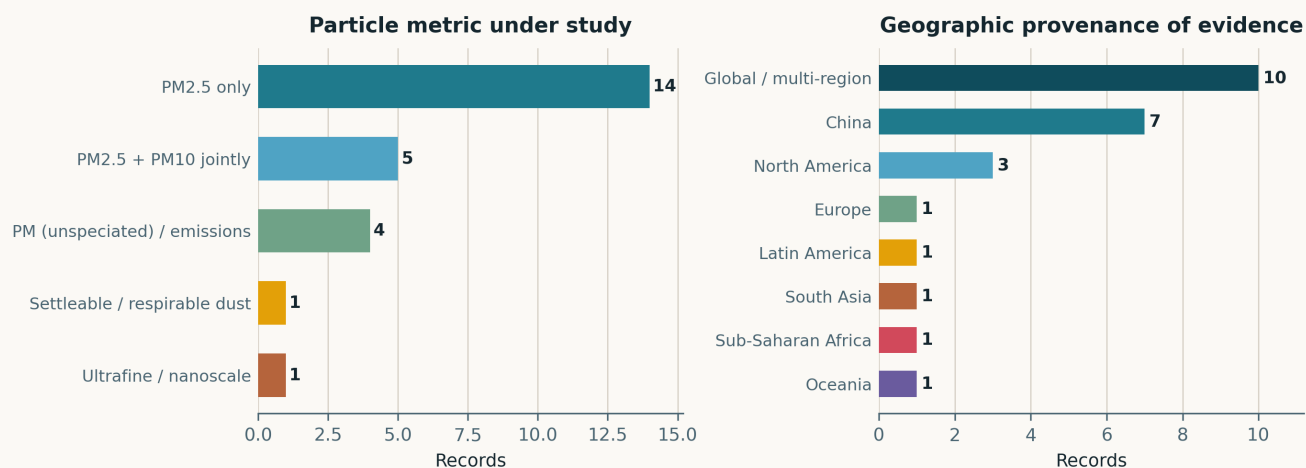
## Corpus analytics



**Fig.** Subtopic assignment of the 25 in-scope records. *Sensing, forecasting & instrumentation* is today's largest cluster (6) — an artefact of methodology, not of the field: PubMed indexed exactly one instrumentation record in the 29 July entry window, and five more arrive from the Consensus sweep that exists to cover journals PubMed does not index. Read the instrumentation count as a backfill, not a daily flux.

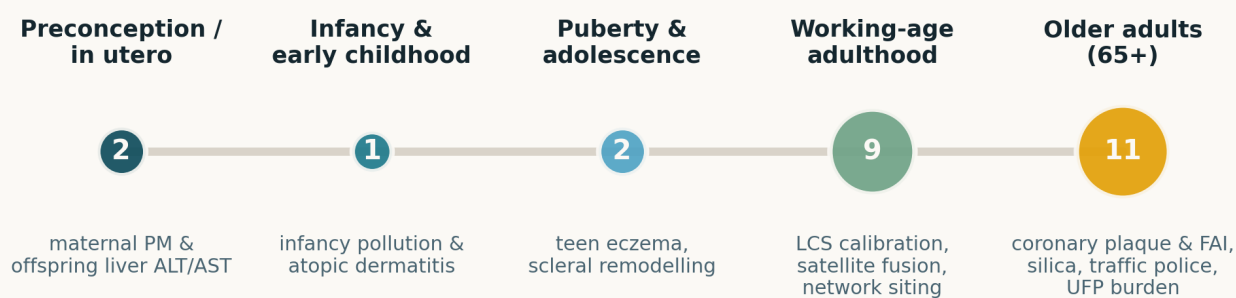


**Fig.** Left — study architecture. Modelling/inventory and observational-cohort work tie at 5 records each; reviews are 4/25 (16%), lower than the 24% of 27 July. Right — health endpoint. Nine records carry **no human health endpoint at all** — five sensing, three exposure-modelling and the risk-based AQI paper. That gap is the structural feature of this issue: instrumentation advances and health inference are still being published into separate literatures with no shared validation target.

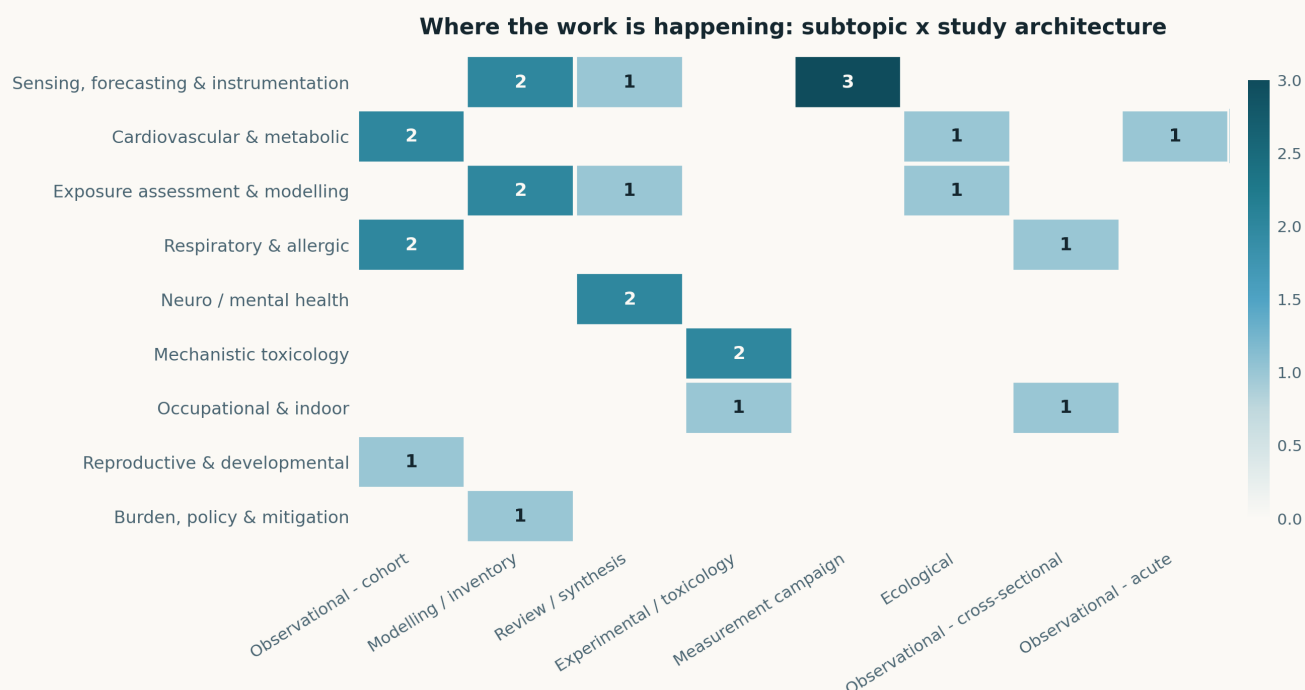


**Fig.** Left — particle metric. PM<sub>2.5</sub> alone is the exposure in 14 records, jointly with PM<sub>10</sub> in 5; ultrafine particles appear once (an ecological country-level analysis, the weakest design in the corpus), continuing the mismatch between deposition physics and measurement practice. Right — geographic provenance. China contributes 7; the *Global/multi-region* block (10) is inflated by the instrumentation backfill, whose papers are method-general. South Asia and sub-Saharan Africa contribute one record each.

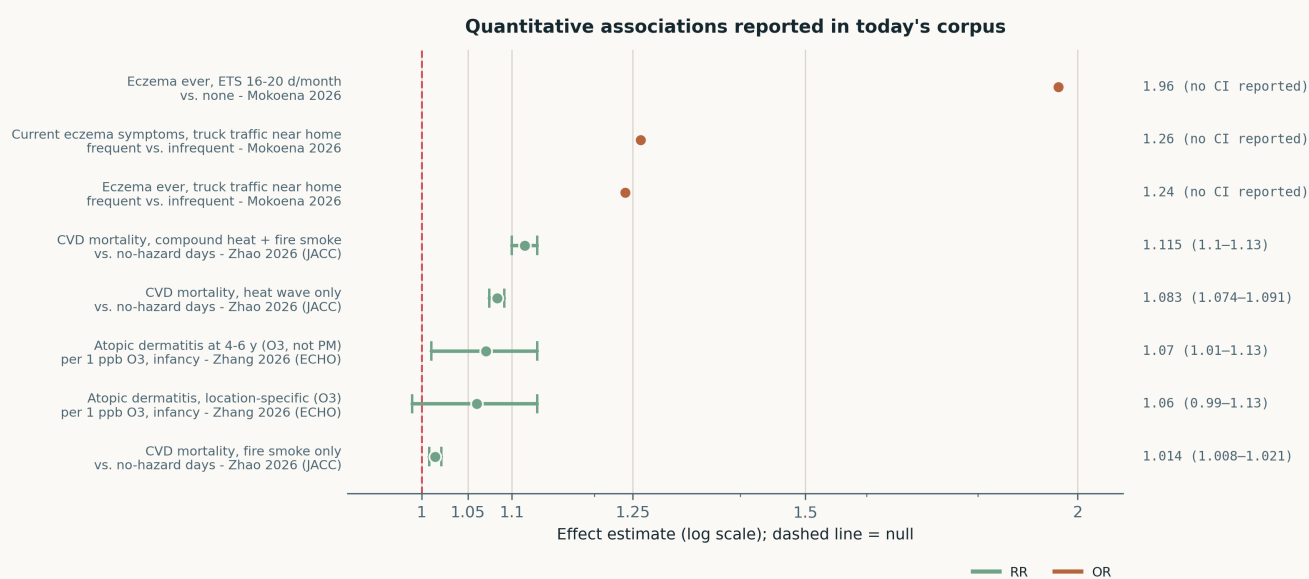
### Life-course windows addressed today (records may span several stages)



**Fig.** Life-course windows addressed. Counts are non-exclusive. Adulthood/ageing dominates (11) on the strength of the coronary-imaging and occupational records; the instrumentation block (9) carries no specific window and is plotted for completeness. Only two records address the in-utero window today, against five on 27 July.



**Fig.** Subtopic × study-architecture cross-tabulation. The diagnostic cell is the intersection of the *Sensing* and *Exposure assessment* rows with any observational-epidemiology column: it is empty. Mechanistic toxicology is again wholly experimental with no human biomarker bridge. Both patterns have now held across all three issues in the corpus.



**Fig.** All quantitative effect estimates in today's corpus on a common logarithmic axis. **Exposure contrasts are not mutually comparable** and the plot shows direction and precision only. The three Zhao et al. estimates share a common referent (no-hazard days) and *are* comparable to each other — that triplet is the informative comparison here. The two ECHO-PATHWAYS estimates are per 1 ppb O<sub>3</sub>, retained because the accompanying PM<sub>2.5</sub> null is itself the finding. Three Mokoena et al. odds ratios are plotted without whiskers because the source reports no confidence intervals; they are self-reported cross-sectional associations and are the weakest evidence shown.

## Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

### Sensing, forecasting & instrumentation

6 records

**Ginsburg et al. 2026** Aerosol Sci Technol

*Global and regional calibration of low-cost PM<sub>2.5</sub> sensors (PurpleAir)*

1,013 PurpleAir units co-located within 500 m of reference monitors on six continents, Jan 2023–Mar 2025, assessed at 1-h and 24-h averaging by R<sup>2</sup>, RMSE, nRMSE, precision (CV) and AQI category correctness. Raw readings overestimate PM<sub>2.5</sub> in most regions and miss EPA performance targets; **US-derived corrections show limited transferability outside North America**; a data-driven global quadratic helps in Australia/NZ and hurts elsewhere; regional equations meet EPA 24-h thresholds in multiple regions and lift 24-h AQI correctness to  $\geq 77\%$ . Limitation: co-location proximity ( $\leq 500$  m) is a proxy for comparability, so some regional “bias” is spatial-gradient error, and continents are represented very unevenly. Still the strongest available empirical case that LCS network calibration must be local.

doi: 10.1080/02786826.2026.2676293

**Feng et al. 2026** Meas Sci Technol

*Calibration strategies and measurement frameworks for low-cost PM sensors*

Review spanning ambient, mobile, indoor and workplace deployments (community networks, vehicle-mounted platforms, smart-manufacturing floors, mining). Compares purely data-driven, purely physics-based and hybrid physics-informed ML calibration on accuracy, transferability, compute cost and interpretability: non-linear models reach sub-10  $\mu\text{g m}^{-3}$  error but resist embedded deployment and depend on representative training data; no strategy is universally optimal. Names the persistent barriers plainly — sensor fouling, data governance, model generalisation, hardware limits — and points to self-cleaning optics, federated learning and knowledge-distilled on-device calibration. Narrative rather than quantitative synthesis, with no pooled performance estimate, so it frames rather than settles the question.

doi: 10.1088/1361-6501/ae6298

**Patton et al. 2026** Environ Sci Pollut Res

*Human health risk of particle exposure in outdoor environments, southwestern Pennsylvania*

ROCIS low-cost monitoring across **290 homes in Allegheny County, 2016–2021**; optical particle counts  $>0.5 \mu\text{m}$  converted to PM<sub>2.5</sub>-equivalent mass, census tracts classified by environmental-justice status and industrial proximity, GEE models plus non-cancer hazard quotients. **33.8% of sites exceeded the WHO 24-h  $15 \mu\text{g m}^{-3}$  guideline and 66.6% the EPA annual  $9 \mu\text{g m}^{-3}$  reference** on 3-week site medians; winter concentrations significantly elevated. The count-to-mass conversion is the weak link — it assumes a fixed density and size distribution that industrial plumes violate — and the authors correctly flag the Sewickley extreme (median  $19.3 \mu\text{g m}^{-3}$ ) as resting on three homes. Directly relevant as a template for community-scale LCS deployment with explicit EJ stratification.

doi: 10.1038/s41598-019-44409-3

**Patel et al. 2026** Proc AAAI

*Scalable air-quality sensor placement via gradient-based mutual information maximization*

Treats sensor coordinates as differentiable parameters and maximises mutual information by continuous optimisation, so runtime is **independent of both the number of placements and candidate-grid density** — the two terms that make greedy MI intractable at national scale. Beats random placement and the maximum-predictive-variance heuristic on a continental PM<sub>2.5</sub> dataset and approaches greedy MI regionally, orders of magnitude faster. Caveat: MI is computed under an assumed GP covariance, so placements inherit that model’s stationarity assumptions, which is exactly where real PM fields misbehave near sources. Immediately useful for network-expansion planning under a fixed sensor budget.

doi: 10.1609/aaai.v40i46.41257

**Huda et al. 2026** J Air Waste Manag Assoc

*Low-cost sensor network for wildfire monitoring at remote community locations*

Solar-powered microsystem (PM<sub>2.5</sub>, CO, CO<sub>2</sub>, O<sub>3</sub>, tVOC) with a microcontroller scheduler and mesh radio, built for forest deployment in northwestern Alberta where there is no cellular or Wi-Fi coverage. Reported PM<sub>2.5</sub> **RMSE  $< 5 \mu\text{g m}^{-3}$** ; mesh range validated at 1–5 km through urban structures and vegetated terrain;  $50 \times 37$  cm panel sized for treetop mounting with energy-aware duty cycling. This is an engineering report, not an evaluation: no co-location period, drift characterisation or seasonal uptime statistics are given, and winter operation is described as prospective. Useful for the power and comms architecture rather than the data-quality claim.

doi: 10.1080/10962247.2026.2665375

**Msigwa et al. 2026** IEEE Internet Things J

*AIoT real-time sensing and calibration of low-cost PM sensors (TCM-Net)*

Customised PMS7003 units with externally PWM-controlled fans to stabilise airflow, on a oneM2M-compliant IoT platform, calibrated by a transformer + temporal-convolutional network with a memory-aware module and trained against a co-located certified reference. Reported **RMSE  $0.325 \mu\text{g m}^{-3}$ , R<sup>2</sup> 0.999**. Those figures sit below the reference instrument’s own measurement uncertainty and cannot be a valid generalisation estimate; the near-certain explanation is temporally adjacent train/test splits on autocorrelated series rather than held-out deployment. The forced-airflow hardware idea is worth taking; the performance number is not.

doi: 10.1109/jiot.2025.3631631

### Exposure assessment & modelling

4 records

**Ding et al. 2026** Geo-spat Inf Sci

*Urban–rural balanced estimation framework for PM<sub>2.5</sub> exposure across China*

Targets a structural bias rather than a model-fit problem: because ground stations cluster in cities, AOD-to-PM<sub>2.5</sub> retrievals systematically overestimate rural concentrations. Integrates AOD, reanalysis fields and ground measurements under data-assimilation principles, adaptively constraining non-urban estimates with low-weighted reanalysis samples, then ensemble-averages across dynamically weighted background/observation settings. China 2015–2021: **clean-site RMSE  $18.18 \rightarrow 8.31 \mu\text{g m}^{-3}$ , national mean PM<sub>2.5</sub> down  $\sim 55\%$** , with corresponding correction of PM-attributable mortality in low-density areas. The 55% figure is the paper’s own counterfactual, validated at clean *sites* that are still station locations — true rural accuracy remains unvalidated by construction. If even partly right, it implies an inherited upward exposure bias across the satellite-based exposure–response literature.

doi: 10.1080/10095020.2026.2624335

**Hu et al. 2026** Geo-spat Inf Sci*Fusion of ground monitoring, satellite retrieval and CTM for seamless hourly 1-km PM<sub>2.5</sub>*

Fills satellite AOD gaps with continuous WRF-Chem AOD, injects physics-informed features (PM<sub>2.5</sub> temporal components, AOD vertical structure) from the CTM, then predicts with a geospatially aware LightGBM that exploits monitoring-site spatial structure. Site-based cross-validation  $R^2$  **0.87**, **RMSE 13.42  $\mu\text{g m}^{-3}$** , **MAE 8.68  $\mu\text{g m}^{-3}$** , with day-and-night coverage demonstrated on one pollution episode. Site-based CV overstates performance away from stations — the standard weakness — and an RMSE of 13.4 is large relative to the WHO annual guideline, so this is useful for episode mapping and weak for long-term exposure assignment. The physics-feature injection is the transferable step.

doi: 10.1080/10095020.2026.2680855

**Sanchez et al. 2026** Indian J Occup Environ Med*Morbidity related to the level-two eruption of Popocatepetl*

Ecological correlational study in communities 15 km from the crater, May 2023, with daily PM<sub>10</sub>/PM<sub>2.5</sub> and **Kriging used to interpolate ash dispersion because the monitoring network is too sparse to resolve it**. 14% of presentations were ash-related, predominantly non-infectious respiratory; Spearman correlations between ash exposure and morbidity were **not statistically significant**. The authors read this as possible biological adaptation, which the design cannot support — a null in a small ecological study with interpolated exposure is far more likely to be measurement error and low power. Its real value here is as a candid worked example of exposure assignment where no formal network exists.

doi: 10.4103/ijocem.ijocem\_146\_24

**Nguyen et al. 2026** Eng Res Express*Satellite-driven PM<sub>2.5</sub> prediction: machine learning and explainable AI*

Systematic review of 2015–2025 work integrating satellite products (LEO and geostationary AOD, top-of-atmosphere reflectance, Ångström exponent) with ML/DL. Finds Random Forest still a robust benchmark, hybrid and transformer architectures better at spatiotemporal dependence, and AOD gap-filling plus normalisation the dominant determinants of model stability; positions XAI as the route to physical plausibility. No pooled quantitative comparison and no protocol registration, so the ranking claims are narrative. Adds little beyond Hu and Ding above; retained as a citation anchor for the physics-guided-learning argument.

doi: 10.1088/2631-8695/ae7028

## Cardiovascular &amp; metabolic

4 records

**Zhao et al. 2026** J Am Coll Cardiol*Compound extreme heat and fire-sourced air pollution vs. cardiovascular mortality*

Multi-country time-series across six countries, **5.9 million CVD deaths**, with each day classified into mutually exclusive states (no hazard / heat-wave only / substantial fire-sourced air pollution only / compound) and quasi-Poisson RRs plus formal additive interaction. **Compound RR 1.115 (1.100–1.130)** vs. heat-only 1.083 (1.074–1.091) and SFAP-only 1.014 (1.008–1.021); RERI 0.018 (0.002–0.035), attributable proportion 1.598% (0.181–3.129). The interaction is statistically present but small — most compound risk is the heat term — and the mutually-exclusive day classification discards dose information within each state. Fire-sourced attribution rests on modelled smoke fields, so exposure misclassification is non-differential but real. Sets the design standard for compound-hazard work.

doi: 10.1016/j.jacc.2026.06.005

**Belay et al. 2026** Eur J Prev Cardiol*Long-term bushfire-related PM<sub>2.5</sub> and coronary plaque burden*

365 asymptomatic adults aged 40–70 with a family history of CAD, **serial coronary CTA at baseline and 3 years**, with daily total and bushfire-attributed PM<sub>2.5</sub> from validated 5 km models assigned by residential postcode. Per 1 µg m<sup>-3</sup> cumulative bushfire PM<sub>2.5</sub>: **total plaque volume +10.14 mm<sup>3</sup> (1.45–18.82) at baseline and +4.92 mm<sup>3</sup> (1.33–8.52) at follow-up**; total PM<sub>2.5</sub> associations were weaker. Note the exposure setting — 10-year means of 5.88 and 5.93 µg m<sup>-3</sup>, i.e. near the WHO guideline — which makes a plaque-volume signal notable but also means the bushfire and total metrics are highly collinear at these levels. Postcode-level assignment in a 365-person sample is the binding limitation.

doi: 10.1093/eurjpc/zwag399

**Chen et al. 2026** Environ Pollut*Long-term PM exposure and coronary inflammation by pericoronary fat attenuation index*

5,528 CCTA patients with a 656-person serial subcohort; FAI extracted for LAD, LCx and RCA by a deep-learning pipeline; individual PM<sub>2.5</sub>, PM<sub>10</sub> and black carbon from a validated 1 km spatiotemporal model; GAM and linear mixed-effects models. Per IQR of 1-year exposure, **RCA FAI rose 7.57 (PM<sub>2.5</sub>), 8.72 (PM<sub>10</sub>) and 2.35 HU (BC)**, with accelerated annual FAI progression longitudinally. FAI is technically sensitive to tube voltage, contrast timing and epicardial adiposity, and a retrospective clinical population undergoing CCTA is selected on suspected disease — so absolute HU shifts should not be read as population effects. The local-inflammation endpoint nonetheless complements Belay's structural one.

doi: 10.1016/j.envpol.2026.128840

**Rusconi et al. 2026** GeoHealth*Long-term ultrafine particle exposure 2010–2019 and cardiovascular burden in high-income countries*

Applies a new global UFP dataset to compute population-weighted exposure for high-income countries and correlates it with country-level CVD DALYs, YLLs and YLDs, reporting “consistent patterns” between long-term UFP exposure and CVD burden. This is a country-level ecological correlation with no individual data, no lag structure and no control for the PM<sub>2.5</sub> with which UFP is strongly collinear — the authors say so, restrict to high-income countries deliberately, and warn against generalisation. It cannot support causal or even population-attributable inference. Included only because it is the corpus's sole ultrafine-focused record, which is itself the point: UFP epidemiology still lacks the exposure infrastructure PM<sub>2.5</sub> has.

doi: 10.1021/acs.est.2c06752

## Respiratory &amp; allergic

3 records

**Zhang et al. 2026** Epidemiology*Infancy air pollution exposure and atopic dermatitis at 4–6 years*

Pooled longitudinal analysis of **1,587 children across six US cities** in the ECHO-PATHWAYS Consortium (2017–2023); first-year-of-life PM<sub>2.5</sub>, NO<sub>2</sub> and O<sub>3</sub> from validated national spatiotemporal models; three caregiver-reported outcomes; Poisson models with prespecified modification by child sex and maternal education. **PM<sub>2.5</sub> and NO<sub>2</sub> null**; O<sub>3</sub> associated with current AD (aRR 1.07 per 1 ppb, 1.01–1.13) and location-specific AD (1.06, 0.99–1.13), evident only in children of lower-educated mothers. Outcome ascertainment is caregiver questionnaire, not clinical examination, and the education-stratified finding is a subgroup result on a modest sample. A well-powered PM<sub>2.5</sub> null against a validated exposure model is worth as much as a positive.

doi: 10.1097/EDE.0000000000002033

**Li et al. 2026** mSystems*Host genetic variation, airway microbiome and environmental exposure vs. respiratory health*

Sputum metagenomes from **1,651 individuals**, with Mendelian randomisation implicating SO<sub>2</sub>, CO and PM<sub>2.5</sub> effects on lung function potentially mediated by *Actinomyces*, *Haemophilus influenzae* and *Veillonella* spp., and genotype-dependent exposure–microbiome interactions (MEOX1 with lower *Filifactor alovis* under pollutant exposure; FAM110D and USP36 with higher *Ralstonia pickettii*/*Neisseria* and lower *Oribacterium*, tracking COPD risk). Fungal loci also reported. This is many hypotheses from one cohort: MR on aggregate exposures with microbial mediators is fragile to pleiotropy and the variant identifiers are rare or novel, so the loci need replication before they mean anything. Mechanistically suggestive for why identical exposure yields divergent lung-function trajectories.

doi: 10.1128/msystems.00442-26

**Mokoena et al. 2026** Public Health Chall*Socio-economic conditions, air pollution sources and eczema symptoms in Soshanguve teenagers*

Cross-sectional modified ISAAC III survey of 2,885 schoolchildren aged 13–14 across 13 schools in Tshwane, South Africa; multiple logistic regression. Current eczema symptoms 26.8%, eczema ever 23.3%. **Frequent truck traffic near the home: AOR 1.26 (current) and 1.24 (ever); ETS 16–20 days/month: AOR 1.96**; female sex, monthly paracetamol use (AOR 2.37) and pet ownership also associated. No confidence intervals are reported for most estimates, exposure is entirely self-reported with no measurement, and the paracetamol association is the textbook reverse-causation trap. Value is geographic coverage — one of very few sub-Saharan records in this corpus — not effect estimation.

doi: 10.3390/ijerph23020182

## Neuro, cognition &amp; mental health

2 records

**Kabir et al. 2026** Health Place*Environmental attributes as moderators of neighbourhood deprivation and mental health*

Systematic review across seven databases plus grey literature, 19 studies included, asking specifically whether *objectively measured* environmental attributes moderate the deprivation–mental-health association. Moderation findings were mixed: greenness and blue space generally protective in disadvantaged neighbourhoods, **air pollution exacerbated poor mental health**, parks and walkability inconsistent. Noise and elder-support centres were each examined in exactly one study. With 19 heterogeneous studies and no meta-analysis, the moderation claim is directional at best, and the authors' call for longitudinal designs is the honest conclusion. Relevant if you intend to use LCS network output as an effect modifier rather than a main exposure.

doi: 10.1016/j.healthplace.2026.103719

**Sarkar et al. 2026** J Appl Toxicol*PM-induced neurotoxicity: neuroinflammation and organelle dysfunction*

Narrative review mapping CNS entry routes (olfactory-bulb translocation, blood–brain barrier disruption, peripheral inflammatory signalling) onto subcellular failure modes: unfolded protein response with CHOP and caspase-12 induction, impaired oxidative phosphorylation, ROS accumulation, mitochondrial permeability transition, and compromised autophagic flux and lysosomal integrity. Coherent as a mechanistic map but adds no new data, and — as with most of this literature — draws heavily on instillation and high-concentration in vitro models whose dosimetry does not correspond to ambient exposure. Use for the organelle-level framing, not for dose–response.

doi: 10.1002/jat.70360

## Mechanistic toxicology

2 records

**Wang et al. 2026** Exp Eye Res*PM exposure, mitochondrial stress and ECM remodelling in scleral fibroblasts*

Combined in vivo and in vitro: topically applied PM produced particle-surrogate distribution to ocular tissues with **scleral thinning and axial elongation**, while human scleral fibroblasts showed raised oxidative stress, mitochondrial ultrastructural abnormality, phenotypic change and altered collagen metabolism; transcriptomics implicated ECM organisation, mitochondrial metabolism and stress-response pathways. The framing as a myopia mechanism is plausible but the exposure route is the problem — *topical* application is not inhalation, and axial elongation in a rodent eye is a weak proxy for human myopia progression. Notable mainly as one of very few records addressing an ocular endpoint at all.

doi: 10.1016/j.exer.2026.111187

**Liu et al. 2026** Environ Toxicol Pharmacol*Resveratrol ameliorates PM<sub>2.5</sub>-aggravated IBS-D via microbiota and metabolite restoration*

Three-arm murine design (IBS-D, PM<sub>2.5</sub>+IBS-D, resveratrol+PM<sub>2.5</sub>+IBS-D) with untargeted metabolomics, 16S microbiota profiling, colonic histopathology, oxidative-stress and barrier-function assays and visceral-hypersensitivity testing. PM<sub>2.5</sub> worsened IBS-D with dysbiosis and disrupted hypoxanthine and linoleic-acid metabolism; resveratrol restored both. No exposure concentrations, particle source or dose–response are reported in the abstract, group sizes are unstated, and the resveratrol arm is a single dose — so this is proof-of-concept, not a therapeutic result. The gut-axis endpoint remains underexplored, which is why it is retained.

doi: 10.1016/j.etap.2026.105120

## Occupational &amp; indoor exposure

2 records

**Qi et al. 2026** Ann Work Expo Health*Emission rates of respirable crystalline silica and dust from engineered stone fabrication*

Extracts emission rates (mass per time) from four published experimental simulation studies of cutting and grinding engineered stone, tests control technology and task as determinants by regression, then drives a **two-zone mathematical model** and validates predictions against task-duration field measurements. RCS emission rate rose with slab crystalline-silica content and fell with control technology; **near-field predictions tended to underestimate measured exposures** though the distributions overlapped. Four source studies is a thin evidence base and the underestimation direction is exactly the wrong way for a protective standard. Methodologically the most transferable record in this cluster — an explicit emission-rate-to-exposure model rather than another exposure survey.

doi: 10.1093/annweh/wxag060

**Ghosh et al. 2026** Indian J Occup Environ Med*Respiratory quality of life and pulmonary function in traffic police, eastern India*

Cross-sectional comparison of 62 traffic police against 120 unexposed controls (Oct 2023–Jan 2024) using the St George's Respiratory Questionnaire and digital spirometry. Exposed personnel had significantly higher SGRQ scores and **lower FVC and FEV<sub>1</sub>** (both  $p < 0.001$ ), with SGRQ correlating positively with pulmonary function measures. No exposure measurement whatsoever — occupation is the exposure proxy — and no adjustment for smoking, duration of service or socioeconomic position is reported, while simple random sampling from unequal groups invites selection differences. Reads as a hazard-surveillance signal for a large occupational group, not as an exposure–response result.

doi: 10.4103/ijoem.ijoem\_150\_25

## Reproductive &amp; developmental

1 record

**Lin et al. 2026** Environ Pollut*Maternal PM exposure during pregnancy and offspring liver function*

Fujian Birth Cohort, **1,046 mother–offspring pairs**, with gestational and trimester-specific PM<sub>1</sub>, PM<sub>2.5</sub> and PM<sub>10</sub>, serum ALT and AST at ages 1–2 years, restricted cubic splines for shape and distributed lag models for timing. Second-trimester IQR increments gave **ALT +1.42 (0.82–2.03), +1.41 (0.77–2.05) and +0.91 (0.35–1.46) U/L** for PM<sub>1</sub>/PM<sub>2.5</sub>/PM<sub>10</sub>, with AST increases of similar magnitude. The effect sizes are small relative to the assay's biological variability in toddlers and clinical significance is unestablished; a single post-natal measurement precludes trajectory inference, and PM<sub>1</sub>/PM<sub>2.5</sub>/PM<sub>10</sub> are too collinear to separate as cleanly as the paper implies. The distributed-lag identification of a

second-trimester window is the durable part.

doi: [10.1016/j.envpol.2026.128835](https://doi.org/10.1016/j.envpol.2026.128835)

### Burden, policy & mitigation

1 record

**Kraus et al. 2026** Int J Hyg Environ Health

*A risk-based air quality index for Germany*

Rebuilds the German AQI on *risk equivalence* rather than convention: pollutant–outcome pairs with causal or likely-causal evidence were identified, effect estimates extracted from systematic reviews, daily-mean effects transformed to hourly using German monitoring data, and equivalence coefficients derived by Monte Carlo simulation **standardised to PM<sub>2.5</sub> as reference**. PM<sub>2.5</sub> thresholds anchor to WHO 2021 and the EU information threshold; PM<sub>10</sub>, NO<sub>2</sub>, O<sub>3</sub> and SO<sub>2</sub> thresholds follow by risk equivalence, with an added multi-pollutant module. Evaluated on 2019 and 2022 hourly station data: a marked shift from “very good” to “good”/“moderate” and modestly more “poor” days. The daily-to-hourly effect transformation is the load-bearing and least verifiable assumption, and the index inherits every bias of the source meta-analyses. The most directly policy-relevant record today, and a clean template for making an index defensible rather than traditional.

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## Corpus register

| First author                                      | Focus                                                                                                                      | Journal                           | Design                       | Metric                                       | Region       |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------|----------------------------------------------|--------------|
| <b>Neuro / mental health</b>                      |                                                                                                                            |                                   |                              |                                              |              |
| Kabir                                             | Neighbourhood socioeconomic inequalities in mental health: moderation by environmental attributes                          | <i>Health Place</i>               | Scoping review               | PM (unspecified) / emissions                 | Global       |
| Sarkar                                            | Particulate-matter-induced neurotoxicity: neuroinflammation and organelle dysfunction                                      | <i>J Appl Toxicol</i>             | Narrative review             | PM <sub>2.5</sub> + PM <sub>10</sub> jointly | Global       |
| <b>Cardiovascular &amp; metabolic</b>             |                                                                                                                            |                                   |                              |                                              |              |
| Belay                                             | Long-term bushfire-related PM <sub>2.5</sub> and coronary plaque burden in asymptomatic adults                             | <i>Eur J Prev Cardiol</i>         | Prospective cohort           | PM <sub>2.5</sub> (bushfire)                 | Australia    |
| Chen                                              | Long-term particulate matter exposure and coronary inflammation by pericoronary fat attenuation index                      | <i>Environ Pollut</i>             | Retrospective cohort         | PM <sub>2.5</sub> + PM <sub>10</sub> jointly | China        |
| Rusconi                                           | Long-term ultrafine particle exposure 2010-2019 in high-income countries and cardiovascular burden                         | <i>GeoHealth</i>                  | Ecological panel             | Ultrafine particles                          | Global       |
| Zhao                                              | Compound exposure to extreme heat and substantial landscape fire-sourced air pollution and cardiovascular mortality        | <i>J Am Coll Cardiol</i>          | Case-crossover               | PM <sub>2.5</sub> (fire-sourced)             | Global       |
| <b>Reproductive &amp; developmental</b>           |                                                                                                                            |                                   |                              |                                              |              |
| Lin                                               | Maternal particulate matter exposure during pregnancy and offspring liver function                                         | <i>Environ Pollut</i>             | Prospective cohort           | PM <sub>2.5</sub> + PM <sub>10</sub> jointly | China        |
| <b>Respiratory &amp; allergic</b>                 |                                                                                                                            |                                   |                              |                                              |              |
| Li                                                | Human genetic variation, airway microbiome, environmental exposure and respiratory health                                  | <i>mSystems</i>                   | Prospective cohort           | PM <sub>2.5</sub> only                       | China        |
| Mokoena                                           | Socio-economic conditions, air pollution sources and eczema symptoms among teenagers in Soshanguve                         | <i>Public Health Chall</i>        | Cross-sectional (multistage) | PM (unspecified) / emissions                 | South Africa |
| Zhang                                             | Air pollution exposure in infancy and incidence of atopic dermatitis in preschool-aged children                            | <i>Epidemiology</i>               | Prospective cohort           | PM <sub>2.5</sub> only                       | USA          |
| <b>Mechanistic toxicology</b>                     |                                                                                                                            |                                   |                              |                                              |              |
| Liu                                               | Resveratrol ameliorates PM <sub>2.5</sub> -aggravated IBS-D via gut bacteria and hypoxanthine/linoleic acid metabolism     | <i>Environ Toxicol Pharmacol</i>  | Murine inhalation exposure   | PM <sub>2.5</sub> only                       | China        |
| Wang                                              | PM exposure, mitochondrial stress and extracellular matrix remodelling in scleral fibroblasts                              | <i>Exp Eye Res</i>                | In vitro + organoid + murine | PM (unspecified) / emissions                 | China        |
| <b>Occupational &amp; indoor</b>                  |                                                                                                                            |                                   |                              |                                              |              |
| Ghosh                                             | Respiratory quality of life and pulmonary function among traffic police in eastern India                                   | <i>Indian J Occup Environ Med</i> | Cross-sectional (multistage) | PM (unspecified) / emissions                 | India        |
| Qi                                                | Emission rates of respirable crystalline silica and dust from engineered stone fabrication                                 | <i>Ann Work Expo Health</i>       | Bench / device experiment    | Settleable / respirable dust                 | Global       |
| <b>Sensing, forecasting &amp; instrumentation</b> |                                                                                                                            |                                   |                              |                                              |              |
| Feng                                              | Calibration strategies and measurement frameworks for low-cost particulate matter sensors: a review                        | <i>Meas Sci Technol</i>           | Critical review              | PM <sub>2.5</sub> only                       | Global       |
| Ginsburg                                          | Global and regional calibration of low-cost PM <sub>2.5</sub> sensors (PurpleAir): transferability of US-based corrections | <i>Aerosol Sci Technol</i>        | Field measurement            | PM <sub>2.5</sub> only                       | Global       |
| Huda                                              | Low-cost sensor network for wildfire monitoring and air quality at remote community locations                              | <i>J Air Waste Manag Assoc</i>    | Field measurement            | PM <sub>2.5</sub> only                       | Canada       |
| Msigwa                                            | AIoT real-time sensing and calibration of low-cost PM sensors using the TCM network                                        | <i>IEEE Internet Things J</i>     | Deep-learning model          | PM <sub>2.5</sub> only                       | Global       |
| Patel                                             | Scalable air-quality sensor placement via gradient-based mutual information maximization                                   | <i>Proc AAAI</i>                  | Deep-learning model          | PM <sub>2.5</sub> only                       | Global       |
| Patton                                            | Human health risk of particle exposure in outdoor environments in southwestern Pennsylvania                                | <i>Environ Sci Pollut Res</i>     | Real-time sensor campaign    | PM <sub>2.5</sub> only                       | USA          |
| <b>Exposure assessment &amp; modelling</b>        |                                                                                                                            |                                   |                              |                                              |              |
| Ding                                              | Urban-rural balanced estimation framework for PM <sub>2.5</sub> exposure assessment across China                           | <i>Geo-spat Inf Sci</i>           | Deep-learning model          | PM <sub>2.5</sub> only                       | China        |

| First author                           | Focus                                                                                                                          | Journal                           | Design                           | Metric                                       | Region  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------|----------------------------------------------|---------|
| Hu                                     | Fusion of ground monitoring, satellite observation and chemical transport modelling for seamless hourly 1-km PM <sub>2.5</sub> | <i>Geo-spat Inf Sci</i>           | Deep-learning model              | PM <sub>2.5</sub> only                       | China   |
| Nguyen                                 | Satellite-driven prediction of PM <sub>2.5</sub> : machine learning and explainable AI                                         | <i>Eng Res Express</i>            | Critical review                  | PM <sub>2.5</sub> only                       | Global  |
| Sanchez                                | Morbidity related to level-two eruption of Popocatepetl in communities near the crater                                         | <i>Indian J Occup Environ Med</i> | Ecological panel                 | PM <sub>2.5</sub> + PM <sub>10</sub> jointly | Mexico  |
| <b>Burden, policy &amp; mitigation</b> |                                                                                                                                |                                   |                                  |                                              |         |
| Kraus                                  | Development and evaluation of a risk-based air quality index for Germany                                                       | <i>Int J Hyg Environ Health</i>   | Bayesian + geospatial regression | PM <sub>2.5</sub> + PM <sub>10</sub> jointly | Germany |

## Provenance and method

**PubMed** (NCBI E-utilities) was queried on 29 July 2026 against [Date - Entry] for 2026/07/29 on two separate axes: particle terms crossed with health-outcome terms (18 hits) and monitoring, sensing and calibration terms crossed with particle terms (1 hit, off-topic). **Europe PMC** CREATION\_DATE for the same day returned 1 record, already present in the PubMed set. **OpenAlex** returned HTTP 429/403 on the created-date filter — that leg remains broken, as recorded on 28 July. The **arXiv** relevance query returned 60 entries, **none submitted on or after 20 July 2026**, so it contributed nothing; its date-range parameter is still unusable. Because PubMed structurally does not index *Atmos. Meas. Tech.*, *Aerosol Sci. Technol.*, *Meas. Sci. Technol.* or the IoT and ML venues where sensor-calibration work appears, **eight** instrumentation and modelling records were added from a **Consensus** sweep (LCS calibration and drift, co-location against reference monitors, satellite–surface fusion, network siting, edge inference). **Consensus offers no entry-date filter**, so those eight are dated by DOI registration (2025-11 to 2026-06) rather than by a 29 July entry window, and are flagged as a *backfill* wherever counts are reported — notably in the subtopic figure. All eight were checked against *state/seen.json* and none had appeared in a previous issue. **3 records excluded**: a lithium-ion-battery ultrasonic-sensing paper (PMID 42525745, matched on “sensor” + “machine learning”, no PM content); a workshop proceedings with no data (PMID 42524492); and one IoT paper whose abstract was unavailable from every source (10.1109/jiot.2026.3707055) — reasons logged to *state/rejected.json*. **Two PubMed records carry demonstrably wrong DOIs in their source metadata**: PMID 42521969 (Pennsylvania LCS study) is registered against a 2019 *Sci Rep* DOI and PMID 42524698 (UFP/CVD) against an *Environ Sci Technol* DOI. Both are reproduced as indexed, so those two links will misresolve — cite by PMID. **Clinical-trial watch**: one registration was created or updated in the 2026-07-23→29 window — NCT05874479, a 440-participant sham-controlled portable air-cleaner trial for blood pressure in New York City public housing (NYU Langone, recruiting, primary completion 2027-09). Accumulated to *state/trials.json* and surfaced in the weekly rollup, not here. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Effect estimates are transcribed verbatim from abstracts and use heterogeneous exposure contrasts — see the forest-plot caption. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.